Automated Tax Classification & Risk Analysis

An Explainable AI Approach using **XGBoost** & NLP

By: Ishan Malik (im2854@nyu.edu)

Prepared for EY Tax Technology Assessment

Made with GAMMA

Executive Summary

I built a robust, audit-ready pipeline that transforms messy client data into accurate tax decisions.

- **Objective:** Automate General Ledger (GL) coding to reduce manual effort.
- **Challenge:** High noise data (corrupted IDs, unstructured text).
- **Solution:** Multi-modal **XGBoost** pipeline with NLP (TF-IDF).

96%

Test Accuracy

XG Boost Model Used

0.92

MCC Score

Reliability

~70%

Auto-Approval

Potential

95.6%

FI-Score

Smaller Data

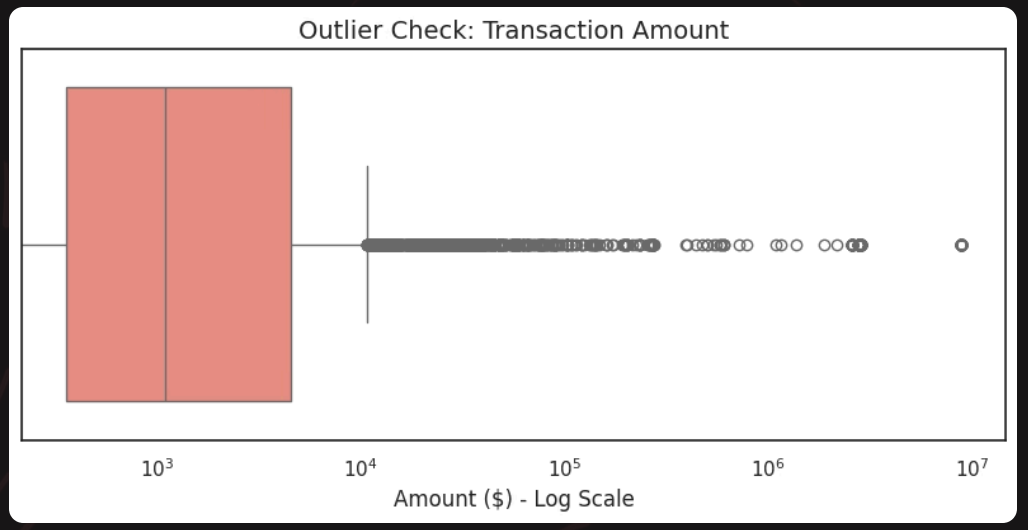
Made with GAMMA

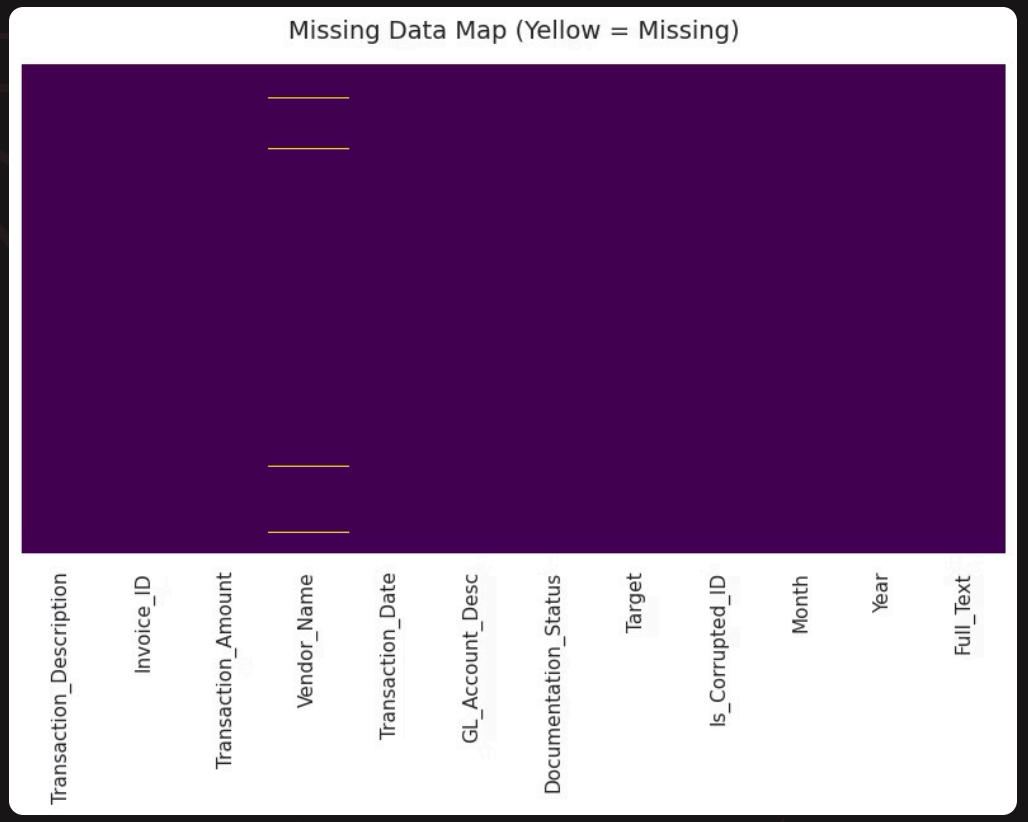
Data Cleaning

The Problem

Client data contained systematic errors: 'Z' typos in Invoice IDs, inconsistent capitalization, and extreme financial outliers.

The Solution: I didn't just model the data; I sanitized it. I engineered a 'Corrupted ID' flag which turned out to be predictive, turning a data quality issue into a feature.





Made with GAMMA

Model Architecture

1

NLP Layer

TF-IDF Vectorization to read Invoice Descriptions & Vendor Names.

2

Feature Engineering

Log-transformation of Amounts, Date extraction (Seasonality), & Quality Flags.

3

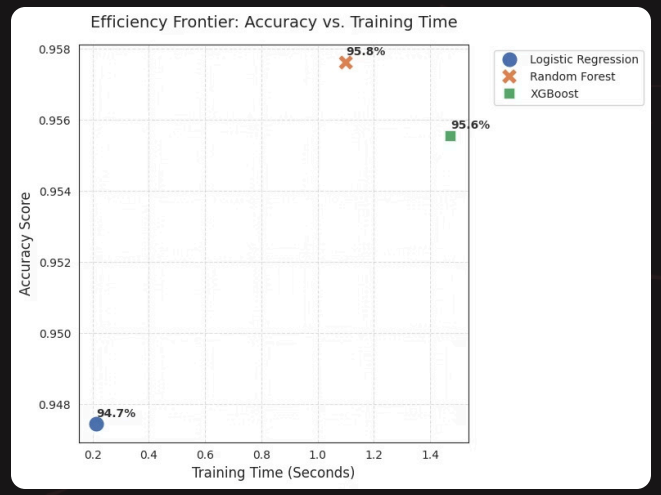
The Engine

Gradient Boosted Decision Trees optimized for Multi-Class LogLoss.

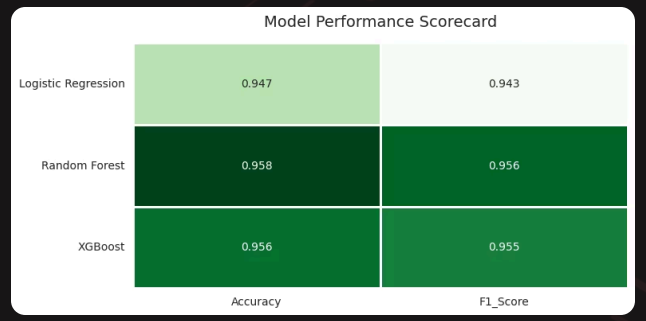
Made with GAMMA

Model Comparison

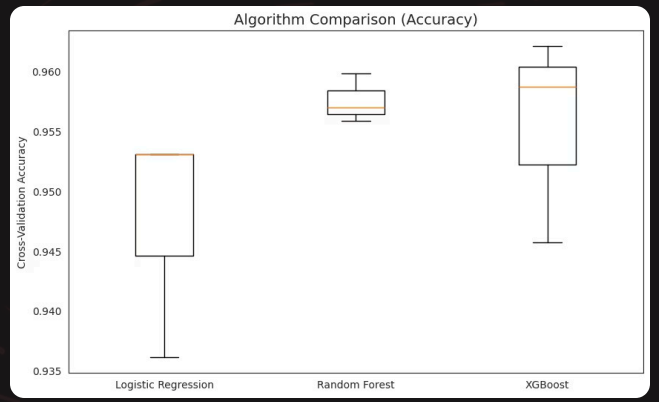
I compared Logistic Regression, Random Forest, and XGBoost. As shown in the scatter plot, XGBoost provided the best trade-off between speed and accuracy. Finally, **XG Boost was the winner** and hence, was used.



Scatter Plot - Accuracy v/s Training Time



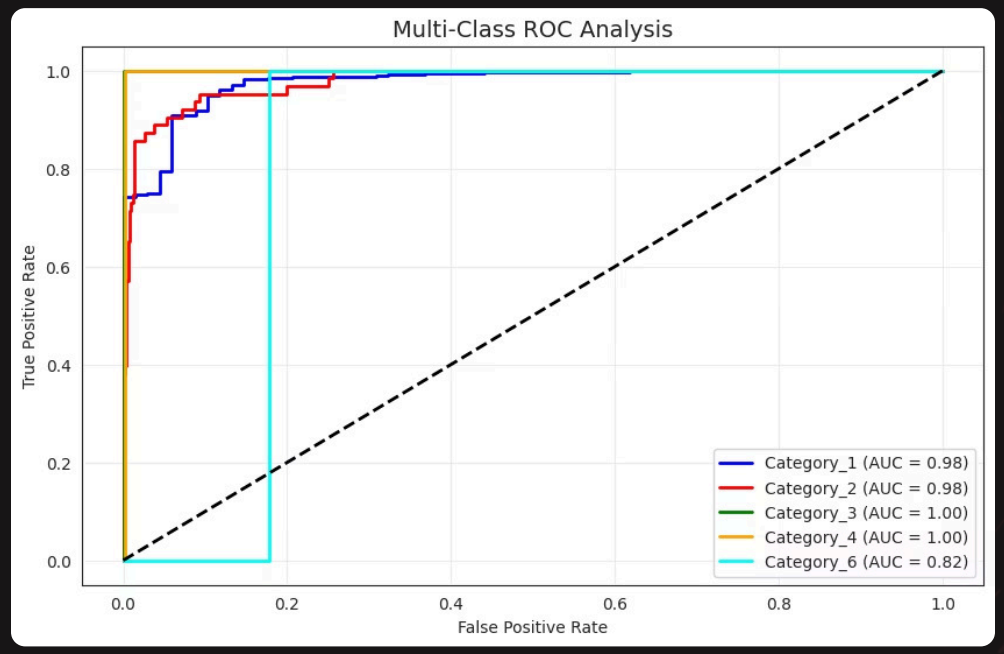
Performance Scorecard

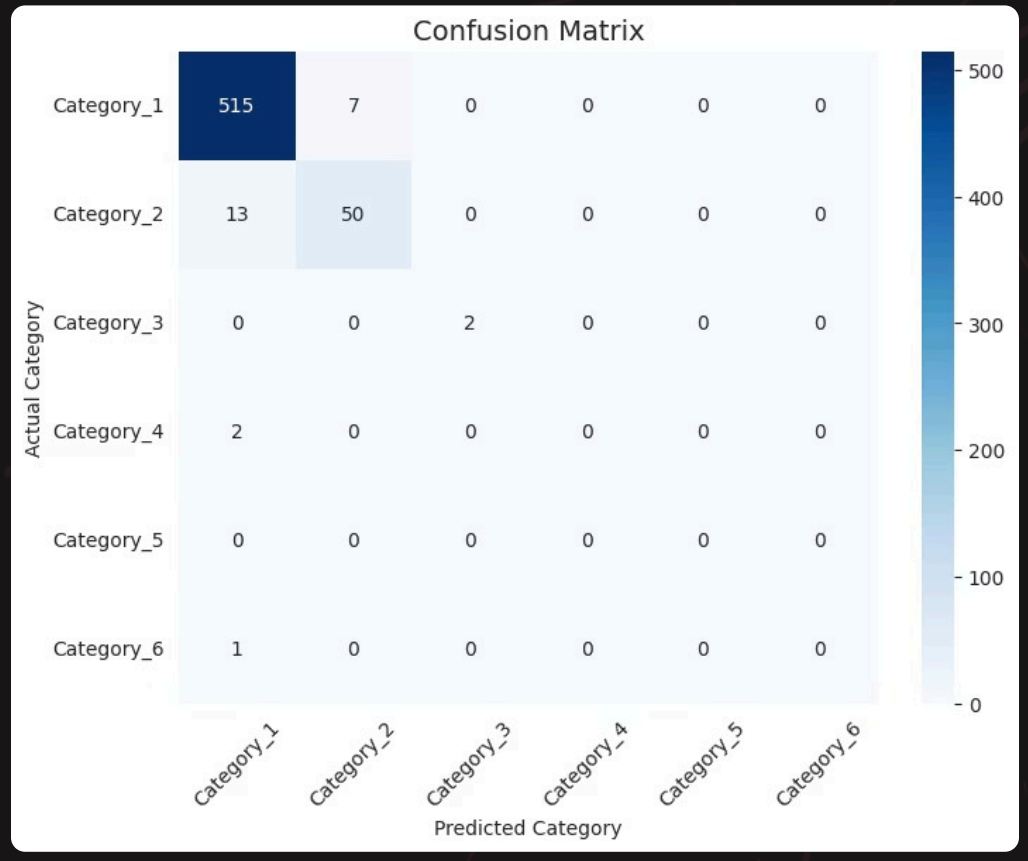


Box Plot for Model-Wise Accuracy

Made with GAMMA

Performance Evaluation





Key Metrics

- **Test Accuracy:** 96%
- **MCC (Matthews Correlation Coefficient):** 0.92.
- **Precision/Recall:** High balance, ensuring low False Positives.

Key Insight: Accuracy on the held-out **10% test set** was 96%. More importantly, the Confusion Matrix shows minimal leakage between high-risk categories like For Example : Between 'Meals' and 'Travel' (Based on my interpretation of the data)

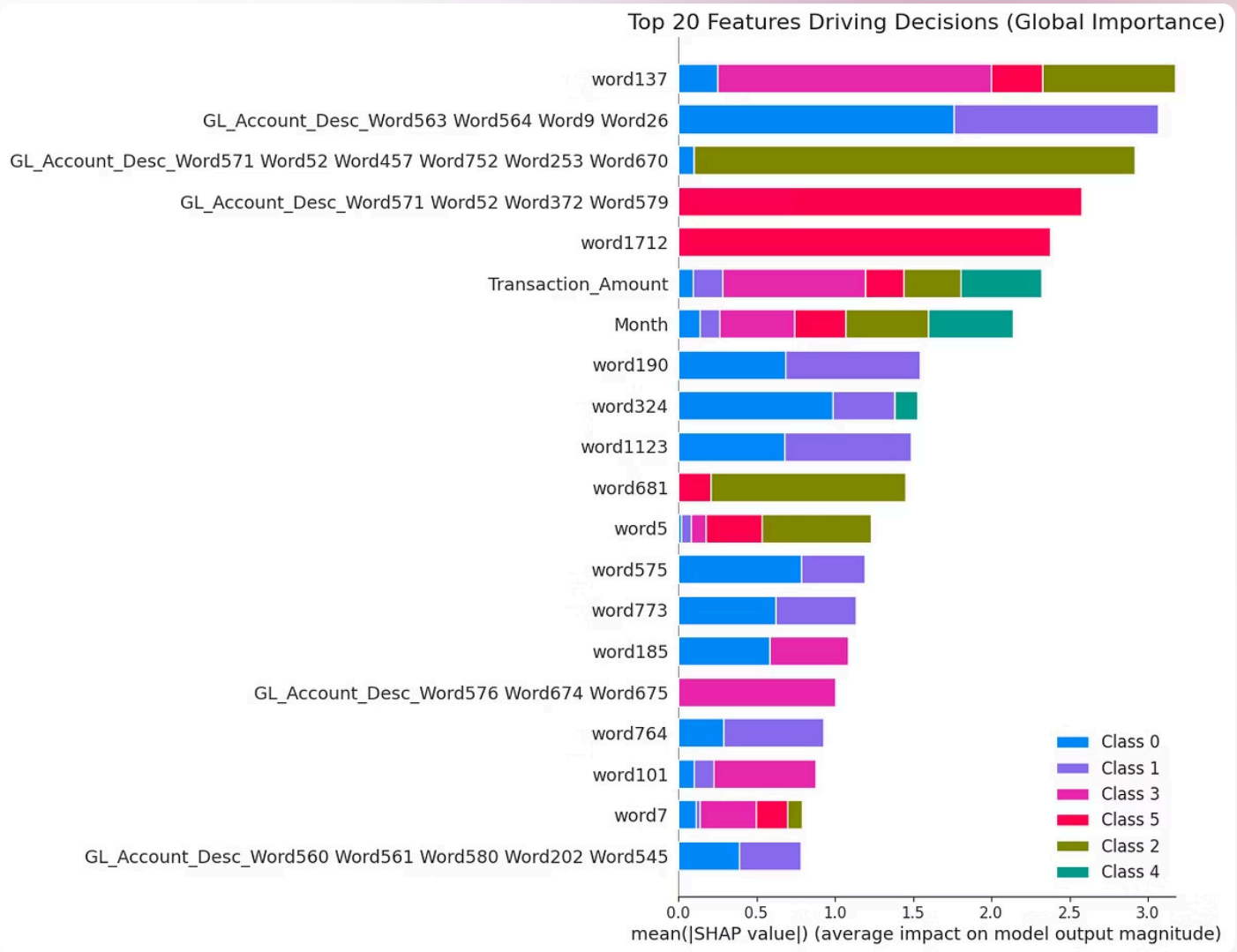
```
--- 1. Detailed Classification Report ---
              precision    recall  f1-score   support

  Category_1       0.97      0.99      0.98       522
  Category_2       0.88      0.79      0.83        63
  Category_3       1.00      1.00      1.00         2
  Category_4       0.00      0.00      0.00         2
  Category_5       0.00      0.00      0.00         0
  Category_6       0.00      0.00      0.00         1

    accuracy                           0.96       590
   macro avg       0.47      0.46      0.47       590
weighted avg       0.96      0.96      0.96       590
```

```
--- 2. Key Performance Indicators (KPIs) ---
Matthews Correlation Coeff (MCC): 0.8009 (1.0 is perfect)
Cohen's Kappa Score:              0.7984 (Inter-rater reliability)
Multi-Class Log Loss:             0.1226 (Lower is better)
```

Made with GAMMA



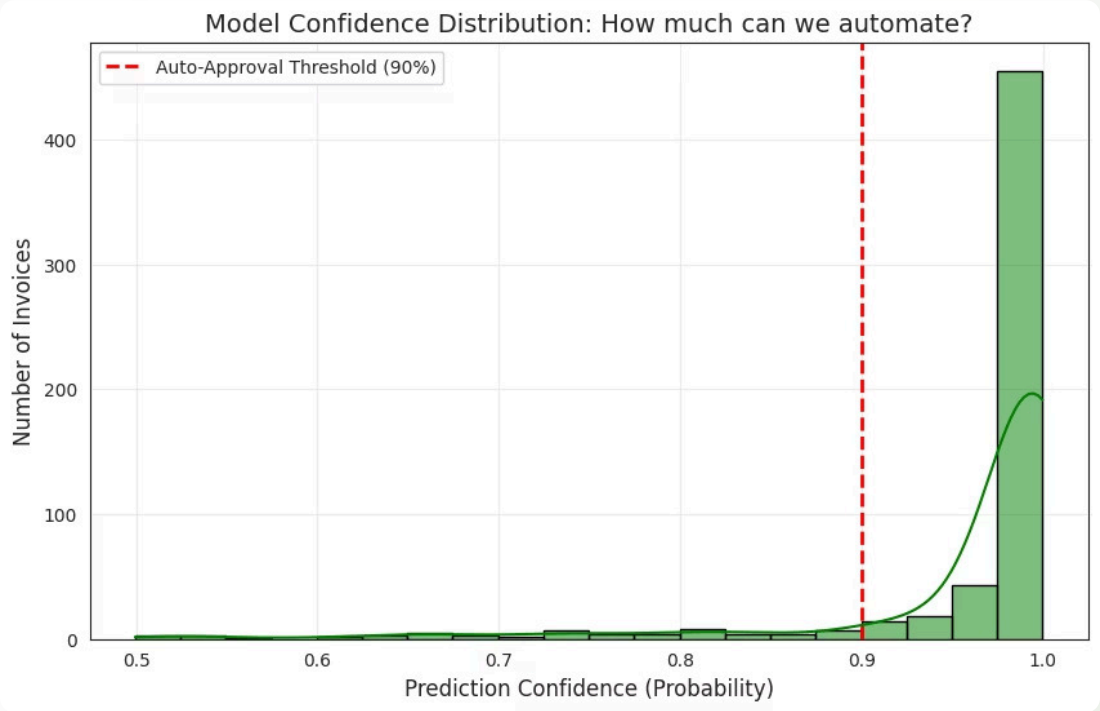
The "White Box" (Explainability)

We can explain why the model made a decision.

Key Insight: While 'Transaction Amount' sets the baseline, specific keywords (NLP) drive the specific category.

For Auditors: A black box is a risk. This SHAP chart proves the model is learning business logic, not noise. You can see that words like For Example: **word137 (say: Software)** and **GL_Account_Desc_Word563..... (say: Uber)** are the top drivers.

Made with GAMMA



Operational Impact

The Strategy

High Confidence (>90%)

Auto-Approve (Direct to GL)

Low Confidence (<90%)

Flag for Human Review

Impact: Reduces manual workload by **approx 70%**.

ROI Strategy: By setting a confidence threshold at 90%, we can safely automate the majority of transactions, allowing our team at EY to focus only on the ambiguous edge cases.

Made with GAMMA

THANK YOU FOR THE CHALLENGE! I LOVED IT.

-By Ishan Malik (im2854@nyu.edu)

(New York University (NYU) - Center for Data Science)

Made with GAMMA